

$$\pi_1 = (2, 1, 0, 1)^T \quad \pi_2 = (1, 0, 1, 2)^T$$

$$f(\pi_1) = (4, 1, 2, 5)^T = 1\pi_1 + 2\pi_2$$

$$f(\pi_2) = (2, 2, -2, -2)^T = 2\pi_1 - 2\pi_2$$

Matricea operatorului $A = \begin{pmatrix} 1 & 2 \\ 2 & -2 \end{pmatrix}$

$$\det \begin{pmatrix} 1-\lambda & 2 \\ 2 & -2-\lambda \end{pmatrix} = 0$$

$$(1-\lambda)(-2-\lambda) - 4 = 0$$

$$-2 - \lambda + 2\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$\lambda_{1,2} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$$

$$\lambda = 2$$

$$A - 2I = \begin{pmatrix} 1-2 & 2 \\ 2 & -2-2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}$$

$$\begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$-x + 2y = 0$$

$$x = 2y$$

$$y = 1 \rightarrow x = 2$$

$$w_1 = (2, 1)$$

$$\lambda = -3$$

$$A + 3I = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \xrightarrow{\cdot (2)}$$

$$2x + y = 0$$

$$y = -2x$$

$$\text{let } x = 1 \Rightarrow y = -2$$

$$w_2 = (1, -2)$$

$$x = (5, 1, 3, 1)^T$$

$$x = a\pi_1 + b\pi_2$$

$$x = a(2, 1, 0, 1) + b(1, 0, 1, 2) = (2a+b, a, b, a+2b)$$

$$2a + b = 5$$

$$a = 1$$

$$b = 3$$

$$a + 2b = 7$$

$$\left. \begin{matrix} a = 1 \\ b = 3 \end{matrix} \right\}$$

$$[x]_B = (1, 3)$$

$$(1, 3) = c_1 w_1 + c_2 w_2 \Rightarrow 1 \cdot w_1 - 1 \cdot w_2$$

$$c_1(2, 1) + c_2(1, -2) = (2c_1 + c_2, c_1 - 2c_2)$$

$$2c_1 + c_2 = 1$$

$$c_1 - 2c_2 = 3$$

$$c_2 = 1 - 2c_1 \quad \wedge \quad c_2 = -1$$

$$c_1 - 2 + 4c_1 = 3$$

$$5c_1 = 5$$

$$c_1 = 1$$

$$A w_1 = 2w_1 \quad A w_2 = -3w_2$$

$$A^n w_1 = 2^n w_1 \quad A^n w_2 = (-3)^n w_2$$

$$A^n(1, 3) = A^n(w_1 - w_2) = 2^n w_1 - (-3)^n w_2$$

Ujmoči možetú úpradu

dostaneme reláciu $2^n(2, 1) - (-3)^n(1, -2)$

pretože $2^{n+1} - (-3)^n \} (k_n, \beta_n) = (2^{n+1} - (-3)^n, 2^n + 2(-3)^n)$

ak $n \sim n$ $2^n + 2(-3)^n$

Príklad de $\mathbb{R}^4$

~~uvažujme~~ $f^n(x) = k_n x_1 + \beta_n x_2$

dostaneme $k_n(2, 1, 0, 1) + \beta_n(1, 0, 1, 2) = (2k_n + \beta_n, k_n, \beta_n, 2k_n + 2\beta_n)$